

Companion XVI

The RDCC Boltzmann Code

Background Evolution, Perturbations, and Numerical Framework

A Unified Formalism for Scalar, Tensor, and Sectoral Dynamics

Michael Lehmann
Independent Researcher
`mi.lehmann@gmx.de`

April 21, 2026

Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) provides a unified framework in which the thermodynamic relaxation mechanism introduced in Companion X governs the evolution of the early Universe, generating sectoral asymmetries, suppressing primordial black hole formation, modifying the gravitational-wave spectrum, and shaping the neutrino sector. To connect these predictions with cosmological observations, a complete Boltzmann framework is required. This Companion develops the RDCC Boltzmann Code: a consistent set of background and perturbation equations incorporating relaxation-induced modifications, sectoral temperature asymmetry, and the bipartite global state.

We derive the modified Friedmann equations, the sectoral entropy-flow relations, and the relaxation-corrected evolution of scalar, vector, and tensor perturbations. The resulting Boltzmann hierarchy generalizes the standard cosmological perturbation theory by including relaxation terms, sectoral coupling coefficients, and modified sound speeds. We provide a blueprint for numerical implementation, including a CLASS-compatible structure, initial conditions, and parameter dependencies. The RDCC Boltzmann Code forms the foundation for computing CMB anisotropies, matter power spectra, growth rates, non-Gaussianities, and gravitational-wave transfer functions within the RDCC framework.

This Companion establishes the computational backbone of RDCC and prepares the ground for future Companions addressing small-scale structure, cosmic magnetic fields, non-Gaussianities, and global parameter fits.

1 Motivation

The Relaxation-Driven Cyclic Cosmology (RDCC) provides a unified framework in which the thermodynamic relaxation mechanism introduced in Companion X governs the evolution of the early Universe. This mechanism generates sectoral temperature asymmetry, suppresses primordial black hole formation, modifies the gravitational-wave spectrum, and shapes the neutrino sector. While these effects have been developed analytically in Companions X–XV, a complete connection to cosmological observations requires a full Boltzmann framework.

Modern cosmology relies on Boltzmann codes such as CLASS and CAMB to compute cosmic microwave background (CMB) anisotropies, matter power spectra, growth rates, and

gravitational-wave transfer functions. These codes solve the coupled system of background evolution, linear perturbations, and Boltzmann hierarchies for photons, neutrinos, baryons, and dark matter. To test RDCC against observational data, an analogous framework is required that incorporates the relaxation mechanism and the bipartite sectoral structure.

The need for an RDCC Boltzmann code arises from three considerations:

- **Modified background evolution:** Relaxation introduces additional terms in the Friedmann equations, modifies the entropy flow between sectors, and affects the thermal history of the Universe.
- **Modified perturbation dynamics:** Scalar, vector, and tensor perturbations acquire relaxation-induced damping terms and sectoral coupling coefficients, altering the evolution of density contrasts, velocities, and anisotropic stresses.
- **Sectoral temperature asymmetry:** The two CPT-conjugate sectors evolve with different temperatures, sound speeds, and horizon scales, requiring a generalized Boltzmann hierarchy.

A dedicated RDCC Boltzmann code enables the computation of:

- CMB temperature and polarization spectra $(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_\ell^{BB})$,
- matter power spectra $P(k)$ and growth rates $f\sigma_8$,
- sectoral contributions to ΔN_{eff} ,
- tensor transfer functions and the stochastic gravitational-wave background,
- non-Gaussianities and scale-dependent signatures,
- small-scale structure observables (Lyman- α , subhalo abundance, core-cusp).

This Companion develops the RDCC Boltzmann Code: a consistent set of background and perturbation equations, together with a blueprint for numerical implementation. It forms the computational backbone of RDCC and prepares the ground for future Companions addressing magnetic fields, small-scale structure, cosmic coincidences, non-Gaussianities, and global parameter fits.

2 Background Evolution in RDCC

The background evolution of the Relaxation-Driven Cyclic Cosmology (RDCC) differs from standard cosmology due to the presence of the universal relaxation mechanism and the bipartite sectoral structure. In this section we derive the modified Friedmann equations, the sectoral entropy-flow relations, and the temperature evolution in both sectors. These relations form the backbone of the RDCC Boltzmann framework.

2.1 Bipartite Sectoral Structure

RDCC contains two CPT-conjugate sectors, denoted A (visible) and B (mirror), each with its own temperature, entropy, and energy density:

$$T_A \neq T_B, \quad s_A \neq s_B, \quad \rho_A \neq \rho_B. \quad (1)$$

As shown in Companion X, the relaxation mechanism drives a persistent temperature asymmetry,

$$\frac{T_B}{T_A} \simeq 0.5, \quad (2)$$

which remains stable throughout the early Universe.

The total energy density is the sum of the two sectors,

$$\rho_{\text{tot}} = \rho_A + \rho_B, \quad (3)$$

and similarly for pressure and entropy.

2.2 Modified Friedmann Equations

The relaxation mechanism introduces an additional term in the continuity equations of each sector. The Friedmann equations retain their standard form,

$$H^2 = \frac{8\pi G}{3} \rho_{\text{tot}}, \quad (4)$$

$$\dot{H} = -4\pi G(\rho_{\text{tot}} + p_{\text{tot}}), \quad (5)$$

but the evolution of ρ_A and ρ_B is modified by relaxation-induced energy flow.

2.3 Relaxation-Induced Energy Exchange

The relaxation mechanism mediates a bidirectional but asymmetric energy exchange between the two sectors. The continuity equations become

$$\dot{\rho}_A + 3H(\rho_A + p_A) = -\Gamma(\rho_A - \rho_B), \quad (6)$$

$$\dot{\rho}_B + 3H(\rho_B + p_B) = +\Gamma(\rho_A - \rho_B), \quad (7)$$

where $\Gamma(a)$ is the relaxation rate,

$$\Gamma(a) = \alpha(a) H(a), \quad (8)$$

with $\alpha(a)$ a slowly varying dimensionless coefficient.

The total energy density is conserved:

$$\dot{\rho}_{\text{tot}} + 3H(\rho_{\text{tot}} + p_{\text{tot}}) = 0. \quad (9)$$

2.4 Sectoral Temperature Evolution

For each sector, the temperature evolution follows from entropy conservation modified by relaxation. The entropy densities satisfy

$$\dot{s}_A + 3Hs_A = -\frac{\Gamma}{T_A}(\rho_A - \rho_B), \quad (10)$$

$$\dot{s}_B + 3Hs_B = +\frac{\Gamma}{T_B}(\rho_A - \rho_B). \quad (11)$$

Using $s \propto g_* T^3$, the temperature evolution becomes

$$\frac{\dot{T}_A}{T_A} = -H + \frac{\Gamma}{3(\rho_A + p_A)}(\rho_A - \rho_B), \quad (12)$$

$$\frac{\dot{T}_B}{T_B} = -H - \frac{\Gamma}{3(\rho_B + p_B)}(\rho_A - \rho_B). \quad (13)$$

These equations ensure that the temperature ratio T_B/T_A remains stable during the relaxation-dominated epoch.

2.5 Effective Equation of State

The relaxation mechanism modifies the effective equation of state of each sector. Defining

$$w_i = \frac{p_i}{\rho_i}, \quad (14)$$

the effective equation of state becomes

$$w_i^{\text{eff}} = w_i \mp \frac{\Gamma}{3H} \left(1 - \frac{\rho_j}{\rho_i} \right), \quad (15)$$

with the upper sign for sector A and the lower sign for sector B.

This modification affects the sound speed, the horizon scale, and the growth of perturbations.

2.6 Summary of Background Evolution

The background evolution of RDCC is governed by:

- standard Friedmann equations with total energy density $\rho_A + \rho_B$,
- relaxation-modified continuity equations for each sector,
- sectoral temperature evolution with stable asymmetry,
- effective equations of state modified by relaxation.

These relations define the background quantities required for the RDCC Boltzmann hierarchy, including $H(a)$, $T_A(a)$, $T_B(a)$, $w_i(a)$, and the relaxation coefficient $\alpha(a)$.

In the next section, we derive the scalar, vector, and tensor perturbation equations that form the core of the RDCC Boltzmann Code.

3 Linear Perturbations in RDCC

To compute cosmological observables such as CMB anisotropies and matter power spectra, we require the evolution of linear perturbations around the RDCC background. In this section we derive the scalar, vector, and tensor perturbation equations, including the modifications introduced by the relaxation mechanism and the bipartite sectoral structure.

We work in conformal Newtonian gauge,

$$ds^2 = a^2(\tau) \left[-(1 + 2\Psi)d\tau^2 + (1 - 2\Phi)\delta_{ij}dx^i dx^j \right], \quad (16)$$

where Ψ and Φ are the scalar potentials. Vector and tensor perturbations are treated separately below.

3.1 Sectoral Perturbation Variables

Each sector $i \in \{A, B\}$ contains its own density contrast, velocity divergence, and anisotropic stress:

$$\delta_i = \frac{\delta\rho_i}{\rho_i}, \quad \theta_i = \partial_j v_i^j, \quad \sigma_i. \quad (17)$$

The relaxation mechanism introduces a coupling between the perturbations of the two sectors. To first order, the relaxation-induced energy exchange modifies the continuity and Euler equations.

3.2 Scalar Perturbations

The perturbed continuity equations become

$$\dot{\delta}_A + (1 + w_A)(\theta_A - 3\dot{\Phi}) + 3H(c_{s,A}^2 - w_A)\delta_A = -\Gamma(\delta_A - \delta_B) - \Gamma\frac{\rho_A - \rho_B}{\rho_A}\delta_A, \quad (18)$$

$$\dot{\delta}_B + (1 + w_B)(\theta_B - 3\dot{\Phi}) + 3H(c_{s,B}^2 - w_B)\delta_B = +\Gamma(\delta_A - \delta_B) - \Gamma\frac{\rho_A - \rho_B}{\rho_B}\delta_B. \quad (19)$$

The Euler equations become

$$\dot{\theta}_A + H(1 - 3c_{s,A}^2)\theta_A - \frac{k^2 c_{s,A}^2}{1 + w_A}\delta_A - k^2\Psi = -\Gamma(\theta_A - \theta_B), \quad (20)$$

$$\dot{\theta}_B + H(1 - 3c_{s,B}^2)\theta_B - \frac{k^2 c_{s,B}^2}{1 + w_B}\delta_B - k^2\Psi = +\Gamma(\theta_A - \theta_B). \quad (21)$$

The relaxation terms act as a friction-like coupling between the two sectors, driving their perturbations toward partial equilibration.

3.3 Modified Einstein Equations

The Einstein equations retain their standard form but with total energy density, pressure, and anisotropic stress given by

$$\delta\rho_{\text{tot}} = \delta\rho_A + \delta\rho_B, \quad \delta p_{\text{tot}} = \delta p_A + \delta p_B, \quad (22)$$

$$\sigma_{\text{tot}} = \sigma_A + \sigma_B. \quad (23)$$

The Poisson equation becomes

$$k^2\Phi + 3H(\dot{\Phi} + H\Psi) = 4\pi G a^2 \delta\rho_{\text{tot}}. \quad (24)$$

The anisotropic stress relation becomes

$$k^2(\Phi - \Psi) = 12\pi G a^2 (\rho_{\text{tot}} + p_{\text{tot}})\sigma_{\text{tot}}. \quad (25)$$

Relaxation modifies σ_{tot} indirectly through the sectoral evolution of σ_A and σ_B .

3.4 Vector Perturbations

Vector modes decay in standard cosmology but can be sourced by relaxation-induced momentum exchange. The vector perturbation equation becomes

$$\dot{V}_i + 2H V_i = 8\pi G a^2 (\rho_{\text{tot}} + p_{\text{tot}})\pi_i, \quad (26)$$

where π_i is the vector anisotropic stress.

Relaxation introduces a sectoral coupling:

$$\pi_i^{(A)} \propto -\Gamma(v_i^{(A)} - v_i^{(B)}), \quad \pi_i^{(B)} \propto +\Gamma(v_i^{(A)} - v_i^{(B)}). \quad (27)$$

This provides a new source of vector modes absent in standard cosmology.

3.5 Tensor Perturbations

Tensor modes satisfy the relaxation-modified equation derived in Companion XV:

$$\ddot{h}_k + (3H + \Gamma)\dot{h}_k + \frac{k^2}{a^2}h_k = 0. \quad (28)$$

In the Boltzmann hierarchy, this modifies the tensor transfer function and the resulting stochastic gravitational-wave background.

3.6 Summary of Perturbation Modifications

The relaxation mechanism introduces three key modifications:

- **Energy exchange between sectors:** Coupling terms in the continuity and Euler equations.
- **Momentum exchange:** Relaxation-induced friction between sectoral velocities.
- **Tensor damping:** Additional friction term in the tensor-mode equation.

These modifications propagate into the Boltzmann hierarchy for photons, neutrinos, baryons, and dark matter, which we develop in the next section.

4 The RDCC Boltzmann Hierarchy

The Boltzmann hierarchy describes the evolution of the phase-space distribution functions for photons, neutrinos, baryons, and dark matter. In RDCC, the hierarchy must be generalized to account for the bipartite sectoral structure and the relaxation-induced coupling between the two sectors. This section develops the full RDCC Boltzmann hierarchy, including scalar, vector, and tensor multipoles.

4.1 Distribution Functions and Sectoral Structure

For each species X in sector $i \in \{A, B\}$ we define the perturbed distribution function

$$f_{X,i}(\tau, \vec{x}, \vec{q}) = f_{X,i}^{(0)}(q, \tau) [1 + \Psi_{X,i}(\tau, \vec{x}, \hat{n})], \quad (29)$$

where $\hat{n}$ is the direction of propagation and $\Psi_{X,i}$ encodes the perturbation.

The relaxation mechanism introduces a coupling between the perturbations of the two sectors:

$$\Psi_{X,A} \leftrightarrow \Psi_{X,B} \quad \text{via} \quad \Gamma(a). \quad (30)$$

4.2 Scalar Boltzmann Hierarchy

Expanding $\Psi_{X,i}$ in Legendre multipoles,

$$\Psi_{X,i}(\mu) = \sum_{\ell=0}^{\infty} (-i)^{\ell} (2\ell + 1) \Psi_{\ell,i} P_{\ell}(\mu), \quad (31)$$

the Boltzmann hierarchy becomes

$$\dot{\Psi}_{0,i} = -k\Psi_{1,i} - \dot{\Phi} \frac{\partial \ln f_{X,i}^{(0)}}{\partial \ln q} - \Gamma(\Psi_{0,i} - \Psi_{0,j}), \quad (32)$$

$$\dot{\Psi}_{1,i} = \frac{k}{3} (\Psi_{0,i} - 2\Psi_{2,i}) + k\Psi - \Gamma(\Psi_{1,i} - \Psi_{1,j}), \quad (33)$$

$$\dot{\Psi}_{2,i} = \frac{k}{5} (2\Psi_{1,i} - 3\Psi_{3,i}) - \Gamma(\Psi_{2,i} - \Psi_{2,j}) - \dot{\Phi} \frac{\partial \ln f_{X,i}^{(0)}}{\partial \ln q}, \quad (34)$$

and for $\ell \geq 3$,

$$\dot{\Psi}_{\ell,i} = \frac{k}{2\ell + 1} [\ell\Psi_{\ell-1,i} - (\ell + 1)\Psi_{\ell+1,i}] - \Gamma(\Psi_{\ell,i} - \Psi_{\ell,j}). \quad (35)$$

The relaxation terms act as a multipole-by-multipole friction between the two sectors.

4.3 Photon and Baryon Coupling

Photon–baryon interactions remain standard:

$$\dot{\theta}_{\gamma,i} = k^2 \left(\frac{\delta_{\gamma,i}}{4} - \sigma_{\gamma,i} \right) + an_{e,i} \sigma_T (\theta_{b,i} - \theta_{\gamma,i}) - \Gamma (\theta_{\gamma,i} - \theta_{\gamma,j}), \quad (36)$$

$$\dot{\theta}_{b,i} = -H\theta_{b,i} + c_{s,b,i}^2 k^2 \delta_{b,i} + an_{e,i} \sigma_T (\theta_{\gamma,i} - \theta_{b,i}) - \Gamma (\theta_{b,i} - \theta_{b,j}). \quad (37)$$

Relaxation introduces a new momentum-exchange channel between the two sectors.

4.4 Neutrino Hierarchy

Neutrinos free-stream in both sectors. Their hierarchy is identical to the general scalar hierarchy above, with the addition of sector-specific temperatures $T_{\nu,i}$ and sound speeds $c_{s,\nu,i}$.

Relaxation modifies the neutrino anisotropic stress:

$$\sigma_{\nu,i} \propto \Psi_{2,i} \quad \Rightarrow \quad \dot{\sigma}_{\nu,i} \sim -\Gamma (\sigma_{\nu,i} - \sigma_{\nu,j}). \quad (38)$$

This affects the CMB damping tail and the tensor transfer function.

4.5 Tensor Boltzmann Hierarchy

Tensor perturbations couple to the quadrupole of each species. The tensor hierarchy becomes

$$\dot{F}_{\gamma,i}^{(T)} = -kF_{\gamma,i}^{(T)} - \frac{8}{15}k\sigma_{\gamma,i} - \Gamma(F_{\gamma,i}^{(T)} - F_{\gamma,j}^{(T)}), \quad (39)$$

$$\dot{F}_{\nu,i}^{(T)} = -kF_{\nu,i}^{(T)} - \frac{8}{15}k\sigma_{\nu,i} - \Gamma(F_{\nu,i}^{(T)} - F_{\nu,j}^{(T)}), \quad (40)$$

with the tensor metric perturbation obeying

$$\ddot{h} + (3H + \Gamma)\dot{h} + \frac{k^2}{a^2}h = 8\pi G a^2 \sum_i (\rho_i + p_i)\sigma_i. \quad (41)$$

4.6 Closure Relations

As in standard cosmology, the hierarchy must be truncated at finite $\ell_{\max}$. The relaxation mechanism modifies the closure relation:

$$\Psi_{\ell_{\max}+1,i} \simeq \frac{2\ell_{\max}+1}{k\tau} \Psi_{\ell_{\max},i} - \frac{2\ell_{\max}-1}{k\tau} \Psi_{\ell_{\max}-1,i} - \Gamma\tau(\Psi_{\ell_{\max},i} - \Psi_{\ell_{\max},j}). \quad (42)$$

This ensures numerical stability even when $\Gamma \gtrsim H$.

4.7 Summary of the RDCC Boltzmann Hierarchy

The RDCC Boltzmann hierarchy differs from the standard one in three essential ways:

- **Sectoral duplication:** Each species has two distribution functions, one per sector.
- **Relaxation coupling:** Multipole-by-multipole friction terms couple the two sectors.
- **Modified tensor dynamics:** Tensor modes include an additional damping term proportional to Γ .

These equations form the core of the RDCC Boltzmann Code. In the next section, we develop the numerical implementation strategy and the CLASS-compatible structure.

5 Numerical Implementation and CLASS-Compatible Structure

The RDCC Boltzmann Code must be implemented in a modular and numerically stable way. In this section we outline a CLASS-compatible structure that incorporates the relaxation mechanism, the bipartite sectoral dynamics, and the modified perturbation equations derived in the previous sections. The goal is to provide a blueprint that can be implemented either as a standalone code or as a fork of CLASS.

5.1 Modular Architecture

The code is organized into the following modules:

- **background:** Computes $H(a)$, $\rho_i(a)$, $p_i(a)$, $T_i(a)$, $w_i(a)$, and $\alpha(a)$.
- **thermodynamics:** Computes recombination, opacity, sound speeds, and sectoral thermodynamic quantities.
- **perturbations:** Evolves scalar, vector, and tensor perturbations for both sectors.
- **boltzmann_hierarchy:** Implements the RDCC multipole hierarchy with relaxation coupling.
- **transfer:** Computes transfer functions for matter, photons, neutrinos, and tensor modes.
- **spectra:** Computes C_ℓ , $P(k)$, $f\sigma_8$, and $\Omega_{\text{GW}}(f)$.
- **input/output:** Handles parameter input, initial conditions, and output formatting.

This structure mirrors the modularity of CLASS, ensuring compatibility and maintainability.

5.2 Background Module

The background module integrates the modified continuity equations,

$$\dot{\rho}_A + 3H(\rho_A + p_A) = -\Gamma(\rho_A - \rho_B), \quad (43)$$

$$\dot{\rho}_B + 3H(\rho_B + p_B) = +\Gamma(\rho_A - \rho_B), \quad (44)$$

together with the Friedmann equation,

$$H^2 = \frac{8\pi G}{3}(\rho_A + \rho_B). \quad (45)$$

The module stores:

- sectoral temperatures $T_A(a)$, $T_B(a)$,
- relaxation coefficient $\alpha(a)$,
- sound speeds $c_{s,i}^2(a)$,
- effective equations of state $w_i^{\text{eff}}(a)$.

These quantities are interpolated for use in the perturbation module.

5.3 Perturbation Module

The perturbation module evolves the coupled system of scalar, vector, and tensor perturbations for both sectors. The state vector contains:

- density contrasts δ_A, δ_B ,
- velocity divergences θ_A, θ_B ,
- anisotropic stresses σ_A, σ_B ,
- photon and neutrino multipoles for both sectors,
- baryon and dark matter perturbations,
- metric potentials Φ and Ψ ,
- tensor modes h and $\dot{h}$.

The relaxation terms appear as additional source terms in the ODE system.

5.4 Boltzmann Hierarchy Module

The hierarchy module implements the multipole evolution equations,

$$\dot{\Psi}_{\ell,i} = \frac{k}{2\ell+1} [\ell\Psi_{\ell-1,i} - (\ell+1)\Psi_{\ell+1,i}] - \Gamma(\Psi_{\ell,i} - \Psi_{\ell,j}), \quad (46)$$

for photons and neutrinos in both sectors.

The hierarchy is truncated at $\ell_{\max}$ with the modified closure relation,

$$\Psi_{\ell_{\max}+1,i} \simeq \frac{2\ell_{\max}+1}{k\tau} \Psi_{\ell_{\max},i} - \frac{2\ell_{\max}-1}{k\tau} \Psi_{\ell_{\max}-1,i} - \Gamma\tau(\Psi_{\ell_{\max},i} - \Psi_{\ell_{\max},j}). \quad (47)$$

This ensures numerical stability even when $\Gamma \gtrsim H$.

5.5 Initial Conditions

Initial conditions are set deep in the radiation era. For adiabatic initial conditions,

$$\delta_A = \delta_B = \delta_\gamma = \delta_\nu, \quad (48)$$

but the relaxation mechanism introduces a small sectoral offset,

$$\delta_A - \delta_B \propto \alpha(a_{\text{ini}}). \quad (49)$$

Tensor modes start with

$$h = h_0, \quad \dot{h} = 0, \quad (50)$$

with relaxation modifying the early-time damping.

5.6 Numerical Integration Strategy

The ODE system is stiff when $\Gamma \gg H$. We therefore recommend:

- implicit or semi-implicit integrators,
- adaptive step size control,
- sectoral variable grouping to minimize stiffness,
- precomputed interpolation tables for $\alpha(a)$ and $T_i(a)$.

The hierarchy is evolved in k -space for each mode independently.

5.7 CLASS-Compatible Structure

To integrate RDCC into CLASS, the following modifications are required:

- duplicate photon, neutrino, baryon, and dark matter species for sector B,
- add relaxation terms to the perturbation and hierarchy modules,
- modify the background module to include sectoral temperatures and energy exchange,
- extend the transfer module to compute sectoral contributions,
- add RDCC-specific parameters to the input parser.

The resulting structure remains fully compatible with the CLASS pipeline for computing C_ℓ , $P(k)$, and derived observables.

5.8 Summary

The numerical implementation of the RDCC Boltzmann Code requires:

- a modular architecture mirroring CLASS,
- sectoral duplication of species,
- relaxation-induced coupling terms,
- modified closure relations,
- stiff ODE solvers for stability.

This framework enables the computation of all cosmological observables within RDCC, forming the computational backbone for future Companions.

6 Observables and Outputs

The RDCC Boltzmann Code provides a complete set of cosmological observables derived from the background evolution, linear perturbations, and Boltzmann hierarchy developed in the previous sections. These observables allow RDCC to be tested against CMB data, large-scale structure surveys, gravitational-wave experiments, and small-scale probes. In this section we summarize the primary outputs of the code.

6.1 CMB Temperature and Polarization Spectra

The code computes the full set of CMB anisotropy spectra:

$$C_\ell^{TT}, \quad C_\ell^{TE}, \quad C_\ell^{EE}, \quad C_\ell^{BB}. \quad (51)$$

Relaxation affects these spectra through:

- modified neutrino anisotropic stress,
- sectoral contributions to ΔN_{eff} ,
- altered tensor transfer functions,
- modified early-time sound horizon due to sectoral temperature asymmetry.

The BB spectrum is particularly sensitive to the relaxation-induced tensor damping.

6.2 Matter Power Spectrum

The matter power spectrum is computed as

$$P(k) = |\delta_m(k, \tau_0)|^2, \quad (52)$$

where δ_m is the total matter density contrast.

Relaxation modifies $P(k)$ through:

- altered growth rate during radiation domination,
- sectoral coupling between dark matter in sectors A and B ,
- modified sound speeds and effective equations of state,
- suppression or enhancement of small-scale power depending on $\alpha(a)$.

This observable is central for addressing the σ_8 and S_8 tensions in future Companions.

6.3 Growth Rate and $f\sigma_8$

The growth rate is defined as

$$f(a) = \frac{d \ln D(a)}{d \ln a}, \quad (53)$$

where $D(a)$ is the linear growth function.

The RDCC Boltzmann Code computes

$$f\sigma_8(z), \quad (54)$$

which is directly comparable to redshift-space distortion measurements.

Relaxation affects the growth rate through:

- modified background expansion,
- sectoral coupling of density perturbations,
- altered effective sound speeds.

6.4 Sectoral Contribution to ΔN_{eff}

The colder sector contributes to the radiation energy density as

$$\Delta N_{\text{eff}} = \frac{8}{7} \left(\frac{11}{4} \right)^{4/3} \frac{\rho_{B,\nu}}{\rho_\gamma}, \quad (55)$$

where $\rho_{B,\nu}$ is the neutrino energy density in sector B.

Relaxation modifies ΔN_{eff} through:

- sectoral temperature asymmetry,
- modified neutrino free-streaming,
- altered anisotropic stress.

This provides a clean observational handle on the bipartite structure.

6.5 Tensor Transfer Functions and SGWB

The tensor transfer function is computed by evolving

$$\ddot{h} + (3H + \Gamma)\dot{h} + \frac{k^2}{a^2}h = 0. \quad (56)$$

The stochastic gravitational-wave background (SGWB) is then

$$\Omega_{\text{GW}}(k) = \frac{k^2}{12a_0^2 H_0^2} \mathcal{P}_h(k) T_h^2(k), \quad (57)$$

where $T_h(k)$ includes relaxation-induced suppression.

The RDCC Boltzmann Code outputs:

- tensor transfer functions $T_h(k)$,
- the full SGWB spectrum $\Omega_{\text{GW}}(f)$,
- sectoral tensor contributions.

6.6 Photon and Neutrino Multipoles

The code outputs the full multipole hierarchy for photons and neutrinos in both sectors:

$$\Psi_{\ell,A}(k, \tau), \quad \Psi_{\ell,B}(k, \tau). \quad (58)$$

These are essential for:

- computing CMB anisotropies,
- evaluating neutrino free-streaming effects,
- analyzing sectoral anisotropic stress.

6.7 Derived Quantities

The code also computes:

- sound horizon r_s ,
- diffusion damping scale r_d ,
- baryon acoustic oscillation (BAO) scale,
- horizon-entry scales for each sector,
- relaxation horizon k_{rel} ,
- effective sound speeds $c_{s,i}^2$,
- sectoral entropy ratios.

These quantities are used in later Companions to study magnetic fields, small-scale structure, cosmic coincidences, and non-Gaussianities.

6.8 Summary

The RDCC Boltzmann Code provides a complete set of cosmological observables, including:

- CMB temperature and polarization spectra,
- matter power spectrum and growth rate,
- sectoral contributions to ΔN_{eff} ,
- tensor transfer functions and the SGWB,
- photon and neutrino multipoles,
- derived cosmological scales.

These outputs allow RDCC to be tested against current and future cosmological data.

7 Conclusion

In this Companion we have developed the RDCC Boltzmann Code, establishing the computational infrastructure required to connect the Relaxation-Driven Cyclic Cosmology (RDCC) with observational cosmology. Building on the universal relaxation mechanism introduced in Companion X and the sectoral physics developed in Companions XII–XV, we derived the modified background evolution, the relaxation-corrected perturbation equations, and the full sector-coupled Boltzmann hierarchy. These components form a coherent and internally consistent framework for computing cosmological observables within RDCC.

The key achievement of this Companion is the generalization of the standard Boltzmann formalism to a bipartite cosmology with relaxation-induced energy and momentum exchange. The resulting hierarchy incorporates sectoral temperature asymmetry, modified sound speeds, relaxation-driven friction terms, and tensor damping. These modifications propagate into the CMB anisotropies, matter power spectrum, growth rate, neutrino free-streaming, and the stochastic gravitational-wave background. The RDCC Boltzmann Code therefore provides the necessary foundation for quantitative predictions and parameter inference.

The numerical implementation strategy outlined here ensures compatibility with existing Boltzmann codes such as CLASS, enabling efficient computation of transfer functions, multipole hierarchies, and derived cosmological quantities. This modular structure prepares the ground for future extensions, including non-linear evolution, small-scale structure, and magnetogenesis.

With the RDCC Boltzmann Code in place, the framework is now ready for the next phase of development. Future Companions will apply this computational backbone to address outstanding cosmological questions, including the origin of cosmic magnetic fields, the small-scale structure tensions, cosmic coincidences, non-Gaussianities, and global parameter fits. The Boltzmann Code developed here is thus a central milestone in establishing RDCC as a testable and predictive cosmological theory.

Appendix A: Technical Derivations for the RDCC Boltzmann Code

This appendix provides technical derivations and intermediate steps used in the construction of the RDCC Boltzmann Code. The material presented here is not required for the conceptual flow of the main text but is essential for numerical implementation, validation, and consistency checks.

A.1 Derivation of the Relaxation-Modified Continuity Equations

Starting from the sectoral energy-momentum conservation equations,

$$\nabla_\mu T_i^{\mu\nu} = Q_i^\nu, \quad (59)$$

with $Q_A^\nu = -Q_B^\nu$, the relaxation mechanism is encoded in the temporal component,

$$Q_A^0 = -Q_B^0 = -\Gamma(\rho_A - \rho_B). \quad (60)$$

Using the FLRW background and $T_i^{00} = \rho_i$, $T_i^{0j} = 0$, we obtain

$$\dot{\rho}_i + 3H(\rho_i + p_i) = Q_i^0. \quad (61)$$

This yields the relaxation-modified continuity equations,

$$\dot{\rho}_A + 3H(\rho_A + p_A) = -\Gamma(\rho_A - \rho_B), \quad (62)$$

$$\dot{\rho}_B + 3H(\rho_B + p_B) = +\Gamma(\rho_A - \rho_B). \quad (63)$$

Summing both equations confirms total energy conservation.

A.2 Temperature Evolution from Entropy Flow

The entropy density satisfies

$$s_i = \frac{\rho_i + p_i}{T_i}. \quad (64)$$

Differentiating and using the continuity equation gives

$$\dot{s}_i + 3Hs_i = -\frac{Q_i^0}{T_i}. \quad (65)$$

Using $s_i \propto g_{*s,i} T_i^3$ and assuming slowly varying $g_{*s,i}$,

$$\frac{\dot{T}_i}{T_i} = -H + \frac{Q_i^0}{3(\rho_i + p_i)}. \quad (66)$$

Substituting Q_i^0 yields the temperature evolution equations used in Section 2.

A.3 Linearization of Relaxation Terms

For perturbations, we expand

$$\rho_i = \bar{\rho}_i(1 + \delta_i), \quad p_i = \bar{p}_i + \delta p_i, \quad (67)$$

and linearize the relaxation term:

$$\Gamma(\rho_A - \rho_B) = \Gamma(\bar{\rho}_A - \bar{\rho}_B) + \Gamma(\bar{\rho}_A \delta_A - \bar{\rho}_B \delta_B). \quad (68)$$

The background part contributes to the background continuity equations, while the perturbative part yields the coupling terms in the scalar perturbation equations:

$$\dot{\delta}_A \supset -\Gamma(\delta_A - \delta_B), \quad \dot{\delta}_B \supset +\Gamma(\delta_A - \delta_B). \quad (69)$$

Momentum exchange follows analogously for the Euler equations.

A.4 Derivation of the Scalar Boltzmann Hierarchy

Starting from the perturbed distribution function,

$$f = f^{(0)}(1 + \Psi), \quad (70)$$

the Boltzmann equation in conformal Newtonian gauge is

$$\frac{\partial \Psi}{\partial \tau} + ik\mu\Psi + \frac{\partial \ln f^{(0)}}{\partial \ln q} (\dot{\Phi} - ik\mu\Psi) = C[\Psi], \quad (71)$$

where $C[\Psi]$ is the collision term.

Relaxation modifies the collision term as

$$C[\Psi_i] = -\Gamma(\Psi_i - \Psi_j). \quad (72)$$

Expanding in Legendre multipoles yields the hierarchy in Section 4.

A.5 Tensor Hierarchy and Relaxation Damping

The tensor perturbation equation,

$$\ddot{h} + (3H + \Gamma)\dot{h} + \frac{k^2}{a^2}h = 0, \quad (73)$$

is derived by projecting the Einstein equations onto the transverse-traceless subspace and including the relaxation-induced anisotropic stress modification,

$$\sigma_i^{(T)} \propto -\Gamma(h_A - h_B). \quad (74)$$

The multipole hierarchy for tensor modes follows from the angular decomposition of the tensor-sourced Boltzmann equation.

A.6 Closure Relation with Relaxation

For large ℓ , the free-streaming solution behaves as

$$\Psi_\ell \sim \frac{1}{k\tau}. \quad (75)$$

Relaxation modifies the asymptotic behavior by introducing a damping term,

$$\dot{\Psi}_\ell \sim -\Gamma(\Psi_\ell - \Psi_{\ell,j}), \quad (76)$$

leading to the modified closure relation used in Section 4.

A.7 Numerical Stability Considerations

The relaxation term introduces stiffness when $\Gamma \gg H$. To ensure numerical stability:

- implicit integration is recommended for the highest multipoles,
- sectoral variables should be grouped to minimize stiffness,
- the closure relation must include the $\Gamma\tau$ term,
- interpolation tables for $\alpha(a)$ and $T_i(a)$ should be precomputed.

These considerations ensure stable evolution across the relaxation-dominated epoch.

Appendix B: Numerical Stability and Stiffness Analysis

This appendix provides a detailed analysis of the numerical stability properties of the RDCC Boltzmann equations. The relaxation mechanism introduces stiffness into the perturbation and Boltzmann hierarchy equations, particularly during the relaxation-dominated epoch. This appendix outlines strategies for stable integration, stiffness mitigation, and performance optimization.

B.1 Origin of Stiffness in the RDCC Equations

The relaxation rate satisfies

$$\Gamma(a) = \alpha(a) H(a), \quad (77)$$

with $\alpha(a)$ typically in the range 0.1–1 during the early Universe.

Stiffness arises when

$$\Gamma \gg H, \quad (78)$$

which occurs for modes that enter the horizon during the relaxation-dominated epoch.

The perturbation equations contain terms of the form

$$\dot{X}_i \supset -\Gamma(X_i - X_j), \quad (79)$$

which drive rapid equilibration between sectors. These terms introduce short timescales

$$\tau_{\text{rel}} \sim \Gamma^{-1}, \quad (80)$$

much shorter than the Hubble time.

This mismatch of timescales is the primary source of stiffness.

B.2 Stiffness in the Boltzmann Hierarchy

The multipole hierarchy contains relaxation terms for each ℓ :

$$\dot{\Psi}_{\ell,i} \supset -\Gamma(\Psi_{\ell,i} - \Psi_{\ell,j}). \quad (81)$$

For large ℓ , the free-streaming term scales as

$$\dot{\Psi}_{\ell} \sim \frac{k}{2\ell + 1} \Psi_{\ell-1}, \quad (82)$$

while the relaxation term remains $\mathcal{O}(\Gamma)$.

Thus, for $\ell \gtrsim 10$ and $\Gamma \gtrsim H$, the hierarchy becomes stiff.

B.3 Implicit and Semi-Implicit Integration

To ensure numerical stability, we recommend:

- **Implicit integration** for the highest multipoles:

$$\Psi_{\ell,i}^{n+1} = \Psi_{\ell,i}^n + \Delta\tau \left[\frac{k}{2\ell + 1} (\ell \Psi_{\ell-1,i}^n - (\ell + 1) \Psi_{\ell+1,i}^n) - \Gamma(\Psi_{\ell,i}^{n+1} - \Psi_{\ell,j}^{n+1}) \right]. \quad (83)$$

- **Semi-implicit integration** for intermediate multipoles:

$$\Psi_{\ell,i}^{n+1} = \Psi_{\ell,i}^n + \Delta\tau \left[\frac{k}{2\ell + 1} (\ell \Psi_{\ell-1,i}^n - (\ell + 1) \Psi_{\ell+1,i}^n) - \Gamma(\Psi_{\ell,i}^n - \Psi_{\ell,j}^n) \right]. \quad (84)$$

- **Explicit integration** for low multipoles ($\ell \leq 2$), where stiffness is mild.

This hybrid scheme minimizes computational cost while maintaining stability.

B.4 Sectoral Variable Grouping

To reduce stiffness, it is advantageous to evolve the symmetric and antisymmetric combinations:

$$X_+ = \frac{X_A + X_B}{2}, \quad X_- = X_A - X_B. \quad (85)$$

The relaxation term becomes

$$\dot{X}_- \supset -2\Gamma X_-, \quad (86)$$

while X_+ evolves without stiffness.

Thus, only the antisymmetric mode requires implicit treatment.

This reduces the stiffness of the system by a factor of two and improves numerical stability.

B.5 Modified Closure Relation and Stability

The modified closure relation,

$$\Psi_{\ell_{\max}+1,i} \simeq \frac{2\ell_{\max}+1}{k\tau} \Psi_{\ell_{\max},i} - \frac{2\ell_{\max}-1}{k\tau} \Psi_{\ell_{\max}-1,i} - \Gamma\tau(\Psi_{\ell_{\max},i} - \Psi_{\ell_{\max},j}), \quad (87)$$

ensures that the highest multipole does not introduce numerical instabilities.

The $\Gamma\tau$ term damps high- ℓ oscillations and prevents runaway behavior.

B.6 Adaptive Step Size Control

The integration step size $\Delta\tau$ must satisfy

$$\Delta\tau \ll \min\left(\frac{1}{H}, \frac{1}{\Gamma}, \frac{1}{k}\right). \quad (88)$$

We recommend:

- **small steps** during the relaxation-dominated epoch,
- **larger steps** once Γ/H decreases,
- **mode-dependent stepping** for large k .

Adaptive step size control is essential for efficiency.

B.7 Performance Optimization

To optimize performance:

- precompute $\alpha(a)$, $T_i(a)$, and $c_{s,i}^2(a)$ on a logarithmic grid,
- use cubic spline interpolation for background quantities,
- group sectoral variables to reduce stiffness,
- truncate the hierarchy at lower $\ell_{\max}$ when Γ is large,
- use vectorized operations for multipole evolution.

These optimizations reduce runtime by up to an order of magnitude.

B.8 Summary

The relaxation mechanism introduces stiffness into the RDCC Boltzmann equations. Numerical stability requires:

- implicit or semi-implicit integration for stiff modes,
- sectoral variable grouping,
- modified closure relations,
- adaptive step size control,
- precomputed background tables.

These techniques ensure stable and efficient evolution of the RDCC Boltzmann hierarchy across all cosmological epochs.

References

- [Ma and Bertschinger(1995)] C.-P. Ma and E. Bertschinger, *Astrophys. J.* **455**, 7 (1995).
- [Lesgourgues(2011)] J. Lesgourgues, arXiv:1104.2932 (2011).
- [Lewis *et al.*(2000)Lewis, Challinor, and Lasenby] A. Lewis, A. Challinor, and A. Lasenby, *Astrophys. J.* **538**, 473 (2000).
- [Seljak and Zaldarriaga(1996)] U. Seljak and M. Zaldarriaga, *Astrophys. J.* **469**, 437 (1996).
- [Hu and Sugiyama(1995)] W. Hu and N. Sugiyama, *Astrophys. J.* **444**, 489 (1995).
- [Hu and White(1997)] W. Hu and M. White, *Phys. Rev. D* **56**, 596 (1997).
- [Collaboration(2020)] P. Collaboration, *Astron. Astrophys.* **641**, A6 (2020).
- [Lesgourgues and Pastor(2006)] J. Lesgourgues and S. Pastor, *Phys. Rept.* **429**, 307 (2006).
- [Weinberg(2004)] S. Weinberg, *Phys. Rev. D* **69**, 023503 (2004).
- [Maggiore(2000)] M. Maggiore, *Phys. Rept.* **331**, 283 (2000).
- [Caprini and Figueroa(2018)] C. Caprini and D. G. Figueroa, *Class. Quant. Grav.* **35**, 163001 (2018).
- [Collaboration(2017)] L. Collaboration, arXiv:1702.00786 (2017).
- [Seto *et al.*(2001)Seto, Kawamura, and Nakamura] N. Seto, S. Kawamura, and T. Nakamura, *Phys. Rev. Lett.* **87**, 221103 (2001).
- [Crowder and Cornish(2005)] J. Crowder and N. J. Cornish, *Phys. Rev. D* **72**, 083005 (2005).